Yue Samuel Lu

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Research Interests

Galaxy formation/evolution, the circum/inter-galactic medium, non-thermal physics (cosmic rays, magnetic fields), large-scale structures, numerical simulations

Education

Sep 2022 – Present
University of California, San Diego (UCSD)
Ph.D. in *Physics* (expected: 2027-2028)
Thesis: *Non-thermal processes in galaxy formation and evolution*
C.Phil./M.S. in *Physics* (Sep 2024)
GPA: 3.90/4.00
Advisor: *Dušan Kereš*

Sep 2018 – Jun 2022
University of California, Santa Barbara (UCSB)
B.S. in *Physics*; B.S. in *Mathematics*; Minor in *Astronomy*
Overall GPA: 3.82/4.00 (Physics GPA: 3.93, Math GPA: 3.92)
Consecutive Dean's Honors

Positions

2022 – Present
Graduate Student Researcher in FIRE Simulation Project (UCSD)
Studied non-thermal processes in galaxy formation/evolution in FIRE simulations, focusing on cosmic-ray transport and magnetic fields

Academic Student Employee (ASE) (UCSD Physics and A&A departments)
Held multiple teaching positions including instructional/teaching assistants (TA), lab TA coordinator (LTAC), and instructor of records for more than 15 courses

2020 – 2024
Student Researcher on Intergalactic Filaments (KITP, UCSD)
Analyzed structure and dynamics of intergalactic filaments in cosmological simulations; connected to observables (Lyα, X-ray); developed analytic models

2020 – 2022
Undergraduate Researcher on AGN Accretion Disk (UCSB)
Explored MHD simulations run by the `Athena++` code; analyzed an $m = 2$ anomaly in a simulated AGN accretion disk

Selected Research Presentations

Sep 2026
Symposium talk, Galaxy Formation and Evolution in Southern California (GalFRESCA) 2026, UC Irvine

Aug 2026
Conference talk, 2026 Santa Cruz Galaxy Workshop, UC Santa Cruz

May 2026
Invited seminar, Center for Cosmology and AstroParticle Physics (CCAPP), OSU, Columbus, OH
Conference talk, Cosmic Ray Transport Workshop, CCAPP, OSU, Columbus, OH

Mar 2026
Invited seminar, UBC Physics and Astronomy Cosmos Coffee, Vancouver, BC, Canada

Dec 2025
Informal talk, Department of Astronomy, Tsinghua University, Beijing, China
Informal talk, Fudan Center for Astronomy and Astrophysics, Shanghai, China

Sep 2025
Symposium talk, GalFRESCA 2025, UC San Diego

Selected Research Presentations (continued)

Aug 2025	🔖	**Poster**, Cosmic Ecosystems, Perimeter Institute for Theoretical Physics, Waterloo, ON, Canada
Jun 2025	🔖	**Invited presentation**, Max Planck Institute for Extraterrestrial Physics (MPE), online
Sep 2024	🔖	**Symposium talk**, GalFRESCA 2024, Carnegie Observatories, Pasadena, CA
Aug 2024	🔖	**Conference talk** (video link), 2024 Santa Cruz Galaxy Workshop, UC Santa Cruz
Dec 2023	🔖	**Poster**, International Conference on Resolving Galaxy Ecosystems on All Scales, CUHK, Hong Kong, China
Aug 2023	🔖	**Conference talk** (video link), 2023 Santa Cruz Galaxy Workshop, UC Santa Cruz
Sep 2021	🔖	**Symposium talk** (video link), UCSB Undergraduate Physics Research Symposium , online

Publications

As the first author

1. **Y. S. Lu**, D. Kereš, P. F. Hopkins, and S. B. Ponnada, *Magnetic field evolution in feedback-dominated galaxies*, Oct. 2026. arXiv: 2610.17216 [astro-ph.GA].
2. **Y. S. Lu**, D. Kereš, P. F. Hopkins, S. B. Ponnada, C.-A. Faucher-Giguère, and C. B. Hummels, *Constraining cosmic ray transport models using circumgalactic medium properties and observables*, Jan. 2026. 🔗 DOI: 10.1093/mnras/staf1984. arXiv: 2505.13597 [astro-ph.GA].
3. **Y. S. Lu**, N. Mandelker, S. P. Oh, A. Dekel, F. C. van den Bosch, V. Springel, D. Nagai, and F. van de Voort, *The structure and dynamics of massive high-z cosmic-web filaments: three radial zones in filament cross-sections*, Feb. 2024. 🔗 DOI: 10.1093/mnras/stad3779. arXiv: 2306.03966 [astro-ph.CO].

As a major contributor

1. S. B. Ponnada, P. F. Hopkins, **Y. S. Lu**, E. M. Silich, I. S. Butsky, and D. Kereš, *Strong Evidence for Cosmic-Ray-supported ∼L* Galaxy Halos via X-Ray and tSZ Constraints*, Jan. 2026. 🔗 DOI: 10.3847/2041-8213/ae2fd9. arXiv: 2510.13959 [astro-ph.GA].
2. M. Roy, K.-Y. Su, S. Tonnesen, **Y. S. Lu**, C. Hummels, and S. B. Ponnada, *To Survive or To Shatter: The Impact of Cosmic Rays on the Fate of Stripped Cold Clouds*, Jun. 2026. 🔗 DOI: 10.3847/1538-4357/ae664f. arXiv: 2510.21699 [astro-ph.GA].

As a minor contributor

1. K.-Y. Su, G. L. Bryan, P. F. Hopkins, P. Natarajan, S. B. Ponnada, R. Emami, **Y. S. Lu**, and M. Roy, *Modelling cosmic rays at AGN jet-driven shock fronts*, Jan. 2026. 🔗 DOI: 10.1093/mnras/staf2060. arXiv: 2502.00927 [astro-ph.GA].
2. S. B. Ponnada, R. K. Cochrane, P. F. Hopkins, I. S. Butsky, S. Wellons, N. N. Sanchez, C. Hummels, **Y. S. Lu**, D. Kereš, and C. C. Hayward, *Hooks, Lines, and Sinkers: How Active Galactic Nucleus Feedback and Cosmic-Ray Transport Shape the Far-infrared–Radio Correlation of Galaxies*, Feb. 2025. 🔗 DOI: 10.3847/1538-4357/ada280. arXiv: 2410.02971 [astro-ph.GA].

Selected Awards

2025	🔖	**Academic Senate Award**, UCSD
2024	🔖	**Distinguished Junior Teaching Award**, UCSD Physics department
2023	🔖	**Chair's Challenge Award**, UCSD Physics department

Selected Awards (continued)

2022 **Graduation Honor**, UCSB College of Letters and Sciences
Department Honor, UCSB Physics Department

Teaching Experiences

Summer 2026 **Instructor of Record for 1-series lab**, UCSD Physics Department
Administered the lower-division pre-health/biology-major physics lab; managed the LTACs and TAs; piloted on creating new evaluation formats for the course

Fall 2024–Present **Lab TA Coordinator (LTAC) for 1-series lab**, UCSD Physics Department
Oversaw a $\sim$ 1,500-student pre-health/biology-major physics lab; ensured smooth course delivery; trained TAs/graders; managed student feedback

Fall 2022–Present **Teaching Assistant**, UCSD Physics and A&A Department
Led discussion sections and lab sections for undergraduate level physics and astronomy courses; graded homework assignments and/or exams.
Courses taught:

- **PHYS 1-series lab:** introductory lab course designed mainly for pre-health students
- **PHYS 2-series:** general physics aimed for science/engineering majors
- **PHYS 7/ASTR 2:** galaxies and cosmology (general education level)
- **PHYS 13/ASTR 4:** life in the universe (general education level)
- **PHYS 163/ASTR 123:** galaxies (designed for upper division science majors)
- **ASTR 103:** dynamics of radiation and fluid (designed for upper division astronomy majors)
- **ASTR 104:** thermal astrophysics (designed for upper division astronomy majors)

2020–2021 **Math, Physics, and Engineering Tutor**, UCSB Campus Learning Assistance Services (CLAS)
Taught lower division math and physics courses; ran group tutorials and drop-in sessions

2019–2022 **Learning Assistant and Grader**, UCSB Physics Department
Assisted teaching assistants on running physics course discussion sessions; graded assignments and/or exams

Academic service and outreach

2026–2027 **Co-chair for ArXiv Hours Committee**, UCSD A&A Department

2023–2025 **Colloquium and Journal Club Committee Member**, UCSD A&A Department

Apr 2024 **Volunteer**, 2024 Continental U.S. Solar Eclipse Outreach Event

Skills

Human Languages English (advanced), Mandarin Chinese (native), Spanish (elementary)

Coding Languages Python, C/C++, bash, shell, Matlab, Mathematica

Misc. Academic research, teaching, training, consultation, LaTeX typesetting